

$$\begin{aligned}
& \frac{m_2}{2} \rightarrow -C_{\phi^2} + \frac{1}{16\pi^2} \\
& \left(-\frac{1}{4} C_{\phi^2}^2 (g_1^2 + 3 g_2^2) + (-C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} - \overline{a_e^{pr}} (C_{\phi^2} a_e^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr})) LF_{2,1,0} [m_l^p, m_e^r] + \right. \\
& (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} + \overline{a_e^{pr}} (C_{\phi^2} a_e^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr})) LF_{3,1,-1} [m_l^p, m_e^r] + \\
& (-C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr}) - \\
& \overline{a_e^{pr}} (a_e^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr} (C_{\phi 1^2} + C_{\phi^2})) \\
& LF_{3,1,0} [m_l^p, m_e^r] + 3 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr}) + \\
& \overline{a_e^{pr}} (a_e^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr} (C_{\phi 1^2} + C_{\phi^2})) \\
& LF_{4,1,-1} [m_l^p, m_e^r] - 2 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_e^{pr}} (a_e^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr}) + \\
& \overline{a_e^{pr}} (a_e^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_e^{pr} (C_{\phi 1^2} + C_{\phi^2})) LF_{5,1,-2} [m_l^p, m_e^r] - \\
& 3 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (C_{\phi^2} a_d^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr})) LF_{2,1,0} [m_q^p, m_d^r] + \\
& 3 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (C_{\phi^2} a_d^{pr} + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr})) LF_{3,1,-1} [m_q^p, m_d^r] - \\
& 3 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr}) + \\
& \overline{a_d^{pr}} (a_d^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr} (C_{\phi 1^2} + C_{\phi^2})) \\
& LF_{3,1,0} [m_q^p, m_d^r] + 9 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr}) + \\
& \overline{a_d^{pr}} (a_d^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr} (C_{\phi 1^2} + C_{\phi^2})) \\
& LF_{4,1,-1} [m_q^p, m_d^r] - 6 (C_{\phi 2\phi 1} \tilde{\mu} \overline{y_d^{pr}} (a_d^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr}) + \\
& \overline{a_d^{pr}} (a_d^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) + \overline{C_{\phi 2\phi 1}} \tilde{\mu} y_d^{pr} (C_{\phi 1^2} + C_{\phi^2})) LF_{5,1,-2} [m_q^p, m_d^r] - \\
& 3 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + C_{\phi^2} \tilde{\mu} y_u^{pr})) LF_{2,1,0} [m_u^r, m_q^p] + \\
& 3 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + C_{\phi^2} \tilde{\mu} y_u^{pr})) LF_{3,1,-1} [m_u^r, m_q^p] - \\
& 3 (\overline{C_{\phi 2\phi 1}} \overline{a_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + \tilde{\mu} y_u^{pr} (C_{\phi 1^2} + C_{\phi^2})) + \\
& \tilde{\mu} \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \tilde{\mu} y_u^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1})) \\
& LF_{3,1,0} [m_u^r, m_q^p] + 9 (\overline{C_{\phi 2\phi 1}} \overline{a_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + \tilde{\mu} y_u^{pr} (C_{\phi 1^2} + C_{\phi^2})) + \\
& \tilde{\mu} \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \tilde{\mu} y_u^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1})) \\
& LF_{4,1,-1} [m_u^r, m_q^p] - 6 (\overline{C_{\phi 2\phi 1}} \overline{a_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} + \tilde{\mu} y_u^{pr} (C_{\phi 1^2} + C_{\phi^2})) + \\
& \tilde{\mu} \overline{y_u^{pr}} (C_{\phi 2\phi 1} a_u^{pr} (C_{\phi 1^2} + C_{\phi^2}) + \tilde{\mu} y_u^{pr} (C_{\phi^2} + \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1})) LF_{5,1,-2} [m_u^r, m_q^p] + \\
& \frac{3}{2} C_{\phi^2}^2 g_1^2 LF_{2,1,-1} [\tilde{\mu}, m_1] - m_1 \tilde{\mu} g_1^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{2,1,0} [\tilde{\mu}, m_1] - C_{\phi^2}^2 g_1^2 LF_{3,1,-2} [\tilde{\mu}, m_1] + \\
& g_1^2 (2 C_{\phi^2}^2 + 4 \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1} + m_1 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{3,1,-1} [\tilde{\mu}, m_1] - \\
& m_1 \tilde{\mu} g_1^2 (C_{\phi 1^2} + C_{\phi^2}) (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{3,1,0} [\tilde{\mu}, m_1] - \\
& 4 g_1^2 (C_{\phi^2}^2 + 2 \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) LF_{4,1,-2} [\tilde{\mu}, m_1] + \\
& 3 m_1 \tilde{\mu} g_1^2 (C_{\phi 1^2} + C_{\phi^2}) (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{4,1,-1} [\tilde{\mu}, m_1] + \\
& 2 g_1^2 (C_{\phi^2}^2 + 2 \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) LF_{5,1,-3} [\tilde{\mu}, m_1] - \\
& 2 m_1 \tilde{\mu} g_1^2 (C_{\phi 1^2} + C_{\phi^2}) (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{5,1,-2} [\tilde{\mu}, m_1] + \frac{9}{2} C_{\phi^2}^2 g_2^2 LF_{2,1,-1} [\tilde{\mu}, m_2] - \\
& 3 m_2 \tilde{\mu} g_2^2 (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{2,1,0} [\tilde{\mu}, m_2] - 3 C_{\phi^2}^2 g_2^2 LF_{3,1,-2} [\tilde{\mu}, m_2] + \\
& 3 g_2^2 (2 C_{\phi^2}^2 + 4 \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1} + m_2 \tilde{\mu} (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1})) LF_{3,1,-1} [\tilde{\mu}, m_2] - \\
& 3 m_2 \tilde{\mu} g_2^2 (C_{\phi 1^2} + C_{\phi^2}) (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{3,1,0} [\tilde{\mu}, m_2] - \\
& 12 g_2^2 (C_{\phi^2}^2 + 2 \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) LF_{4,1,-2} [\tilde{\mu}, m_2] + \\
& 9 m_2 \tilde{\mu} g_2^2 (C_{\phi 1^2} + C_{\phi^2}) (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{4,1,-1} [\tilde{\mu}, m_2] + \\
& 6 g_2^2 (C_{\phi^2}^2 + 2 \overline{C_{\phi 2\phi 1}} C_{\phi 2\phi 1}) LF_{5,1,-3} [\tilde{\mu}, m_2] - \\
& 6 m_2 \tilde{\mu} g_2^2 (C_{\phi 1^2} + C_{\phi^2}) (\overline{C_{\phi 2\phi 1}} + C_{\phi 2\phi 1}) LF_{5,1,-2} [\tilde{\mu}, m_2] \Big)
\end{aligned}$$